

Electrical and HVAC

Texas Real Estate Commission Standards of Practice · **22 TAC 535.229 and 535.230** · For the TREC state exam (25 questions, pass at 75%)

Two sections, same structure. Electrical splits into service and branch circuits; HVAC splits into heating, cooling, ducts and its own general limitations.

535.229 Electrical Systems

- (a) Service Entrance and Panels
- (b) Branch Circuits, Connected Devices and Fixtures

(a) Service Entrance and Panels — report as Deficient

- A **drop, weatherhead or mast not securely fastened** to the building
- The **absence of, or deficiencies in, the grounding electrode system**
- **Missing or damaged dead fronts or cover plates**
- **Conductors not protected from the edges** of electrical cabinets, gutters or cutout boxes
- Deficiencies in electrical cabinets and panels, clearances, overcurrent devices, bonding and grounding

(b) Branch Circuits — the duty that surprises people

The inspector shall:

- **Manually test the installed and accessible smoke AND carbon monoxide alarms;**
- Report the **type of branch circuit conductors;**
- Report as Deficient the **absence of GFCI protection** in bathroom, garage, outdoor and crawl space receptacles, among other required locations.

Texas makes you actually press the button. Manually testing smoke and CO alarms is an affirmative duty here, not an option. Compare the national exam, where alarms are knowledge questions rather than a required action.

The GFCI list follows the same logic you already learned for the national exam: **water, or at or below grade.** Bathrooms, garages, outdoors, crawl spaces.

535.230 Heating, Ventilation and Air Conditioning

- (a) Heating Equipment
- (b) Cooling Equipment
- (c) Duct Systems
- (d) General Limitations (HVAC)

Heating and cooling both follow the general requirement from 535.227(c): **operate the equipment in at least one mode using ordinary controls at typical settings.** You are not required to dismantle it, calculate its capacity, or determine its efficiency.

Duct systems cover distribution, registers, filters and the condition of the ductwork. The HVAC general limitations exclude things such as capacity and adequacy determinations.

The recurring boundary across both sections: operate and observe, yes; calculate, dismantle or determine adequacy, no. If an option asks you to size or rate something, it is outside the standard.

Check Yourself

1. What must a Texas inspector physically do with smoke and CO alarms?

Manually test the installed and accessible ones.

2. Four GFCI locations named in the rule?

Bathroom, garage, outdoor and crawl space receptacles.

3. How many modes must installed equipment be operated in?

At least one, with ordinary controls at typical settings.

Content follows the official TREC Standards of Practice, 22 TAC 535.227-535.233. Rules are amended periodically — verify against the current version on trec.texas.gov before the exam.